GADAN: Text-Conditioned Coordinate Bias for Remote Sensing Visual Grounding

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Abstract

Remote sensing visual grounding (RSVG) exhibits a pronounced condition-to-search-space asymmetry: a short referring expression provides only a few linguistic constraints, whereas the visual stream exposes thousands of candidate locations. Existing methods largely rely on pretrained language encoders and content-based cross-modal attention, leaving token-to-coordinate compatibility implicit in learned features. We propose Geometry-Aware Spatial Bias Injection (GSBI), a lightweight module that explicitly models this compatibility. GSBI projects contextual token features and normalized visual grid boxes into a shared low-rank space and injects the resulting text-conditioned coordinate bias into pre-softmax cross-attention logits through zero-initialized, input-adaptive per-head coefficients. We integrate GSBI into GADAN, which progressively refines linguistic representations across four visual scales using a compact BiLSTM encoder. On DIOR-RSVG, GADAN achieves 83.90% Pr@0.5, 76.91% Pr@0.7, and 75.11% mIoU. Controlled comparisons show that GSBI improves mIoU by 2.29 points with BiLSTM and by 1.69 points with RoBERTa-base. On OPT-RSVG, GADAN reaches 70.24% mIoU, outperforming LPVA by 4.04 points. Keyword-level analysis further shows substantial gains for direction-sensitive expressions while revealing limitations on ordinal and object-relative relations. These results demonstrate that explicit coordinate conditioning provides an effective spatial inductive bias for accurate RSVG.

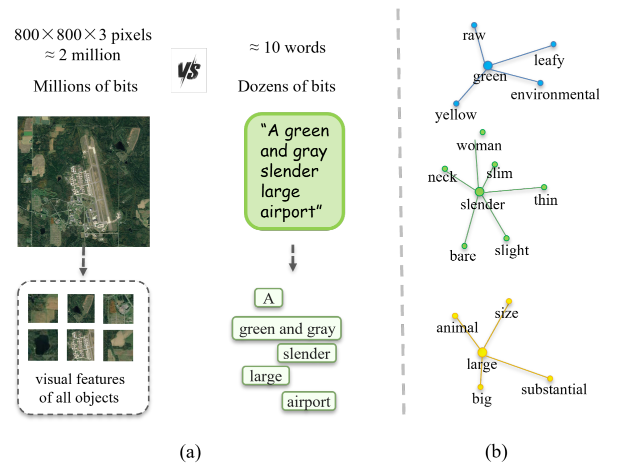

Figure 1: Motivation for GSBI. (a) RSVG exhibits a pronounced information asymmetry at the raw-representation level: millions of visual measurements must be resolved from a query of only a few words.(b) Language pretraining provides broad semantic associations, but image-specific spatial compatibility has to be established against the coordinate layout of the current image. GSBI introduces an explicit text-conditioned coordinate pathway for this purpose.

Introduction

Visual grounding aims to localize the object referred to by a natural-language expression (Hu et al. 2016; Yu et al. 2016, 2018). In remote sensing (RS) imagery, this problem is particularly challenging because aerial scenes cover large spatial extents and contain substantial variations in scale and orientation, dense object distributions, tiny targets, and high inter-class similarity (Li et al. 2020; Xia et al. 2018; Ding et al. 2022a; Xu et al. 2022). Expressions such as "the vehicle on the left of the overpass" and "the ship in the upper-right part of the harbor" therefore require the model to determine both *what* the target is and *where* it is located (Zhan, Xiong, and Yuan 2023; Lan et al. 2024).

A distinctive property of RSVG is the severe information asymmetry between its two input modalities, as illustrated in Figure 1. At the raw representation level, a 640×640 8-bit RGB image contains approximately 9.8×10^6 binary storage units, whereas a typical referring expression contains only a few dozen tokens, corresponding to roughly 10^2–10^3 bits under conventional digital encodings. This comparison concerns raw representational scale rather than Shannon entropy or semantic information content. Nevertheless, it illustrates the strongly asymmetric nature of the grounding problem: a short linguistic condition must disambiguate one target from a dense visual search space (Yuan et al. 2024).

A common response to this asymmetry is to employ a large pretrained language encoder and then fuse the resulting token features with multi-scale visual representations. Such encoders provide useful category, attribute, and contextual knowledge. However, spatial language cannot be resolved from linguistic context alone. The word "left," for example, specifies a relation type, but its realization in a particular sample depends on the coordinate layout and visual context of that image. Standard cross-modal attention can learn this relation implicitly through content features and positional embeddings, but it does not expose a dedicated token-to-coordinate compatibility term. Consequently, the same content projections must simultaneously support appearance matching and location-sensitive selection. This indirect formulation is especially demanding in RS scenes, where several regions may be semantically compatible with the query but differ in spatial suitability.

We address this limitation with **Geometry-Aware Spatial Bias Injection (GSBI)**. For each visual feature level, GSBI maps normalized grid coordinates and contextual text tokens into a compact shared space, producing a text-conditioned coordinate compatibility matrix $\mathbf{B}_{\mathrm{geo}}$. This matrix is added to the semantic attention logits:

$$\mathbf{S}^{(h)} = \frac{\mathbf{Q}^{(h)}(\mathbf{K}^{(h)})^{\top}}{\sqrt{d_h}} + \lambda_{\mathrm{eff}}^{(h)}\mathbf{B}_{\mathrm{geo}}, \tag{1}$$

where the per-head coefficient is zero-initialized and modulated according to the current text and visual features. GSBI therefore preserves standard cross-attention at initialization while allowing the model to learn how coordinate evidence should adjust the content-based attention distribution.

We instantiate GSBI in **GADAN**, a Geometry-Aware Decoupled Attention Network built on the LQVG grounding pipeline (Lan et al. 2024). The same GSBI module is reused across four visual scales, and the text representation is updated sequentially from fine to coarse feature levels. This design introduces a resolution-normalized coordinate pathway without changing the downstream DETR-style grounding objective. GSBI is compatible with both a lightweight randomly initialized BiLSTM and a pretrained RoBERTa-base encoder. Controlled ablations show consistent improvements under either encoder, separating the contribution of coordinate-conditioned fusion from the choice of text representation. We use BiLSTM in the final model because it attains the strongest accuracy under the training protocol.

On DIOR-RSVG, GADAN obtains 83.90% Pr@0.5, 76.91% Pr@0.7, and 75.11% mIoU. Relative to the directly comparable LQVG baseline, the improvements become larger under stricter IoU thresholds, indicating better localization precision. On OPT-RSVG, GADAN surpasses LPVA by 4.04 mIoU points and improves over LQVG by 2.99 mIoU and 2.53 cIoU points, with consistent gains from Pr@0.5 through Pr@0.8. Its Pr@0.9 is 1.87 points lower than LQVG, indicating that the cross-dataset improvement is broad but does not extend to the strictest overlap threshold.

Our contributions are threefold:

1. We revisit RSVG from the perspective of cross-modal information asymmetry and identify a limitation of existing fusion: image-dependent spatial compatibility is represented only indirectly within content-based attention.
2. We propose **GSBI**, which forms a text-conditioned compatibility matrix from normalized visual coordinates and injects it into cross-attention logits using a zero-initialized, input-adaptive per-head coefficient.
3. We develop **GADAN**, which applies GSBI sequentially across multiple visual scales and achieves strong performance on DIOR-RSVG and OPT-RSVG, with gains under both BiLSTM and RoBERTa-base text encoders.

Related Work

Visual Grounding

Early visual grounding systems rank region proposals produced by object detectors using language and contextual features, including spatial configurations and inter-object relations (Ren et al. 2017; Hu et al. 2016; Yu et al. 2016, 2018). One-stage architectures remove the external proposal stage and directly predict the referred region (Yang et al. 2019; Huang et al. 2021). Transformer-based methods further improve cross-modal interaction by representing linguistic and visual elements as tokens and optimizing their correspondence through attention (Carion et al. 2020; Deng et al. 2021; Ye et al. 2022; Yang et al. 2022; Kang et al. 2024). Although these designs differ in query construction, decoding, and verification, their cross-modal compatibility is commonly dominated by learned content projections. Positional features can be included in the visual tokens, yet token-specific compatibility with explicit image coordinates remains implicit. Our work complements content attention with an additional learned coordinate-conditioned logit term.

Representation Learning in Remote Sensing

Remote-sensing deep learning has developed strong visual representations across scene classification, land-cover mapping, hyperspectral analysis, and object detection (Zhang, Zhang, and Du 2016; Zhu et al. 2017). Large-scale aerial benchmarks expose substantial variation in object scale, orientation, scene layout, and visual appearance (Xia et al. 2017, 2018; Li et al. 2020; Ding et al. 2022a). Transfer-learning studies further show that representations pretrained on natural images can support high-resolution remote-sensing recognition, although sensor and geographic shifts remain important (Hu et al. 2015; Tong et al. 2020). Spatial-context models combine convolutional predictions with structured priors, active sampling, multiscale graphs, or spatial–spectral hypergraphs (Cao et al. 2018, 2020; Wan et al. 2020; Luo et al. 2019). These advances provide increasingly capable visual encoders, but they do not by themselves establish token-specific compatibility between language and explicit image coordinates(Kamath et al. 2021; Li et al. 2022).

Remote Sensing Visual Grounding

RSVG has been advanced by benchmarks such as DIOR-RSVG (Zhan, Xiong, and Yuan 2023) and OPT-RSVG (Li et al. 2024). GeoVG (Sun et al. 2022) introduced early task-specific modeling for grounding in aerial scenes, while MGVLF (Zhan, Xiong, and Yuan 2023) explored multi-granularity visual-language fusion. LQVG (Lan et al. 2024) represents referring expressions as language-conditioned queries in a DETR-style architecture and provides the base pipeline used in this work. Recent methods further explore progressive attention, spatial-frequency fusion, scene knowledge, sparse decoding, and text-injected discrimination (Li et al. 2024; Zhao et al. 2024; She et al. 2025; Zhao et al. 2025; Hu, Yang, and Li 2026). Closely related segmentation research has investigated decoupled vision–language recognition, rotated multiscale interaction, and long-term semantic guidance (Ding et al. 2022b; Liu et al. 2024b; Ma et al. 2025). GADAN focuses on a complementary issue: explicitly connecting contextual language tokens with normalized visual coordinates inside the cross-attention logits for bounding-box grounding (Liu et al. 2024a; Zhang et al. 2024b,a; Kuckreja et al. 2024; Liu et al. 2024c).

Position-Aware Attention

Positional information has been incorporated into attention through absolute embeddings, relative position representations, and learned positional biases (Shaw, Uszkoreit, and Vaswani 2018; Raffel et al. 2020; Liu et al. 2021). Most of these mechanisms are designed for self-attention within a single modality and encode distances or offsets between elements of the same sequence. GSBI instead targets cross-modal grounding: contextual text tokens are projected against normalized 2D visual coordinates to obtain a sample-dependent bias over image regions. Unlike a fixed positional prior, the resulting bias changes with the expression and is adaptively weighted for each attention head.

Method

Figure 2 presents the overall pipeline of GADAN. Given an aerial image and a referring expression, GADAN first extracts multi-scale visual features and contextual linguistic features, performs asymmetric bidirectional fusion across four visual levels, and finally predicts the referred bounding box with a DETR-style decoder. The principal modification to the baseline is GSBI in the text-to-visual attention direction; its internal mechanism is illustrated in Fig. 3.

Problem Formulation

Let $\mathbf{I} \in \mathbb{R}^{3\times H_0\times W_0}$ denote an aerial image and $\mathbf{q} = (w_1, \dots, w_M)$ a referring expression of M tokens. The objective of RSVG is to learn a mapping

$$f_{\boldsymbol{\theta}} : (\mathbf{I}, \mathbf{q}) \mapsto \hat{\mathbf{b}} \in [0, 1]^4, \tag{2}$$

where $\hat{\mathbf{b}}$ is the normalized bounding box of the referred target. A visual encoder produces feature maps $\{\mathbf{F}^{(l)}\}_{l=1}^{L}$, which are flattened into visual tokens $\mathbf{H}_{\mathrm{vis}}^{(l)} \in \mathbb{R}^{N_l\times d}$, while the text encoder produces token features $\mathbf{H}_{\mathrm{txt}}^{(0)} \in \mathbb{R}^{M\times d}$. In our implementation, $L = 4$ and $d = 256$.

The central question is how to evaluate the compatibility between token i and visual region j. Standard cross-attention estimates this compatibility entirely from learned content projections. We instead decompose the pre-softmax score into a content term and a coordinate-conditioned term, while preserving the original attention computation as a special case.

Content Affinity and the Missing Coordinate Pathway

For text token i and visual location j, the content-based compatibility in attention head h is written as

$$S_{\mathrm{sem},ij}^{(l,h)} = \frac{\left(\mathbf{h}_{\mathrm{txt},i}^{(l-1)}\mathbf{W}_Q^{(h)}\right)\left(\mathbf{h}_{\mathrm{vis},j}^{(l)}\mathbf{W}_K^{(h)}\right)^{\top}}{\sqrt{d_h}}, \qquad d_h = \frac{d}{H}. \tag{3}$$

Here, $\mathbf{h}_{\mathrm{txt},i}^{(l-1)} \in \mathbb{R}^d$ and $\mathbf{h}_{\mathrm{vis},j}^{(l)} \in \mathbb{R}^d$ denote the representations of the i-th text token and the j-th visual location at feature level l, respectively. The corresponding value features are aggregated after applying softmax over the visual dimension.

Although the content-based logit in Eq. (3) may absorb spatial cues through learned visual and linguistic representations, it does not explicitly parameterize the compatibility between a language token and the coordinates of a visual location. Consequently, location-dependent evidence must be inferred indirectly through the same projections used for appearance-based matching. This limitation is particularly relevant to RSVG, where short referring expressions frequently impose spatial constraints over dense and repetitive visual regions. GSBI therefore retains the content-based semantic logit while introducing a dedicated text-conditioned coordinate term, as detailed in the following subsection.

Geometry-Aware Spatial Bias Injection

Low-Rank Text–Coordinate Compatibility For a feature map of size $H_l \times W_l$, the cell at row r and column c is represented by the normalized box

$$\mathbf{c}_{r,c}^{(l)} = \left[\frac{c}{W_l}, \frac{r}{H_l}, \frac{c+1}{W_l}, \frac{r+1}{H_l}\right]. \tag{4}$$

Stacking all cells gives $\mathbf{C}^{(l)} \in [0, 1]^{N_l\times 4}$. The same coordinate function can therefore be reused at different feature resolutions: cells occupying the same normalized region receive the same coordinate descriptor, irrespective of the absolute feature-map size.

The visual coordinates and contextual text features are lifted into a shared d_{geo}-dimensional space:

$$\mathbf{E}_{\mathrm{vis}}^{(l)} = \mathrm{GELU}\Big(\mathbf{C}^{(l)}\mathbf{W}_1 + \mathbf{b}_1\Big)\,\mathbf{W}_2 + \mathbf{b}_2, \tag{5}$$

$$\mathbf{E}_{\mathrm{txt}}^{(l)} = \mathbf{H}_{\mathrm{txt}}^{(l-1)}\mathbf{W}_{\mathrm{txt}}^{\mathrm{geo}}, \tag{6}$$

where $\mathbf{E}_{\mathrm{vis}}^{(l)} \in \mathbb{R}^{N_l\times d_{\mathrm{geo}}}$ and $\mathbf{E}_{\mathrm{txt}}^{(l)} \in \mathbb{R}^{M\times d_{\mathrm{geo}}}$. Their bilinear compatibility is

$$\mathbf{B}_{\mathrm{geo}}^{(l)} = \mathbf{E}_{\mathrm{txt}}^{(l)}\left(\mathbf{E}_{\mathrm{vis}}^{(l)}\right)^{\top} \in \mathbb{R}^{M\times N_l}. \tag{7}$$

This factorization imposes the structural constraint

$$\mathrm{rank}\Big(\mathbf{B}_{\mathrm{geo}}^{(l)}\Big) \le d_{\mathrm{geo}}, \tag{8}$$

so GSBI learns a compact coordinate compatibility subspace instead of an unconstrained $M \times N_l$ bias table. Because one factor is obtained from contextual text tokens, $\mathbf{B}_{\mathrm{geo}}^{(l)}$ is a *text-conditioned coordinate bias*, rather than a language-independent geometric map. It directly represents image-centric patterns such as left/right, upper/lower, center/corner, and boundary proximity. Explicit pairwise geometry between detected objects is not assumed in the current formulation.

Factorized Adaptive Injection For sample b and attention head h, GSBI factorizes the effective coefficient into

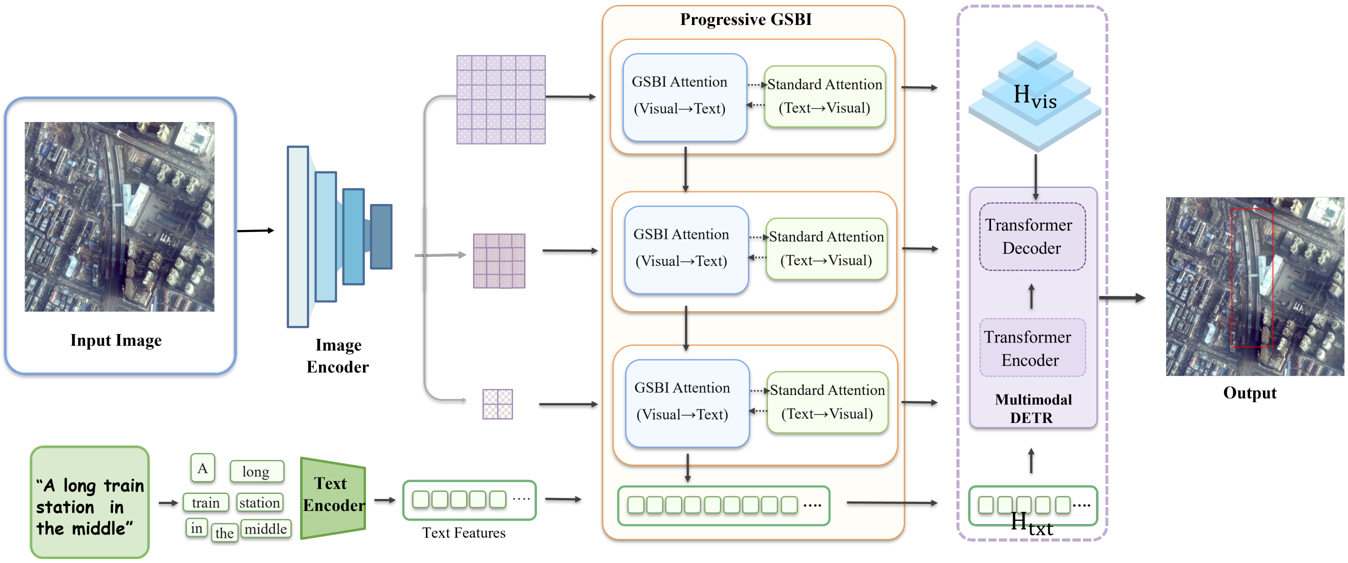

Figure 2: Overall framework of GADAN. The visual and text encoders first extract multi-scale image representations and token-level linguistic features. Bidirectional cross-modal fusion is then performed at successive visual levels, progressively updating the text stream and producing language-conditioned visual features. Finally, the pooled sentence representation and fused multi-scale features are delivered to a Deformable DETR-style decoder for referred-object localization. The internal mechanism of GSBI is detailed in Fig. 3.

a global head-specific parameter and an instance-dependent modulation:

$$\mathbf{z}_b^{(l)} = \left[\bar{\mathbf{h}}_{\mathrm{txt},b}^{(l)}; \bar{\mathbf{h}}_{\mathrm{vis},b}^{(l)}\right], \tag{9}$$

$$g_{b,h}^{(l)} = \sigma\Big(\mathrm{MLP}_{\mathrm{gate}}(\mathbf{z}_b^{(l)})_h\Big), \tag{10}$$

$$\lambda_{b,h}^{(l)} = \lambda_{\mathrm{base}}^{(h)} g_{b,h}^{(l)}, \qquad g_{b,h}^{(l)} \in (0,1), \tag{11}$$

where the bars denote mean pooling over tokens or visual locations. The combined logits are

$$\mathbf{S}_b^{(l,h)} = \mathbf{S}_{\mathrm{sem},b}^{(l,h)} + \lambda_{b,h}^{(l)} \mathbf{B}_{\mathrm{geo},b}^{(l)}. \tag{12}$$

The base coefficient captures the overall geometric propensity of head h, whereas $g_{b,h}^{(l)}$ adjusts that propensity according to the current image–expression pair.

Equation 12 also admits a log-linear interpretation. For token i and location j,

$$A_{ij}^{(l,h)} = \frac{\exp(S_{\mathrm{sem},ij}^{(l,h)}) \exp(\lambda_{b,h}^{(l)} B_{\mathrm{geo},ij}^{(l)})}{\sum_{k=1}^{N_l} \exp(S_{\mathrm{sem},ik}^{(l,h)}) \exp(\lambda_{b,h}^{(l)} B_{\mathrm{geo},ik}^{(l)})}. \tag{13}$$

Thus, GSBI does not replace semantic matching; it multiplicatively reweights semantic affinity in probability space according to the learned coordinate compatibility before normalization.

Initialization consistency. We initialize $\lambda_{\mathrm{base}}^{(h)} = 0$. Consequently,

$$\mathbf{S}^{(l,h)}\Big|_{\lambda_{\mathrm{base}}^{(h)}=0} = \mathbf{S}_{\mathrm{sem}}^{(l,h)}, \tag{14}$$

and GSBI algebraically reduces to standard dot-product cross-attention at initialization. Let $\boldsymbol{\theta}_{\mathrm{geo}}$ denote the parameters of the coordinate and text geometry projections. Their gradient satisfies

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}_{\mathrm{geo}}} = \sum_h \lambda_{b,h}^{(l)} \left\langle \frac{\partial \mathcal{L}}{\partial \mathbf{S}^{(l,h)}}, \frac{\partial \mathbf{B}_{\mathrm{geo}}^{(l)}}{\partial \boldsymbol{\theta}_{\mathrm{geo}}} \right\rangle, \tag{15}$$

which is zero for the first update when the base coefficient is zero. In contrast,

$$\frac{\partial \mathcal{L}}{\partial \lambda_{\mathrm{base}}^{(h)}} = g_{b,h}^{(l)} \left\langle \frac{\partial \mathcal{L}}{\partial \mathbf{S}^{(l,h)}}, \mathbf{B}_{\mathrm{geo}}^{(l)} \right\rangle \tag{16}$$

can be nonzero, allowing optimization to determine whether and in which direction the coordinate pathway should be activated. This provides a controlled initialization without claiming a separate training stage or auxiliary geometric supervision.

After softmax aggregation, the heads are concatenated and projected to obtain $\widetilde{\mathbf{H}}_{\mathrm{txt}}^{(l)}$. Following the fusion operator of the underlying grounding architecture, the text stream is updated by element-wise modulation:

$$\mathbf{H}_{\mathrm{txt}}^{(l)} = \mathbf{H}_{\mathrm{txt}}^{(l-1)} \odot \widetilde{\mathbf{H}}_{\mathrm{txt}}^{(l)}. \tag{17}$$

Multi-Scale Sequential Fusion

The visual encoder provides three backbone feature maps, and a fourth coarser level is generated by a stride-2 3×3 convolution. One GSBI module is shared across all levels. The resulting recursion is

$$\mathbf{H}_{\mathrm{txt}}^{(l)} = \mathrm{GSBI}\Big(\mathbf{H}_{\mathrm{txt}}^{(l-1)}, \mathbf{H}_{\mathrm{vis}}^{(l)}, \mathbf{C}^{(l)}\Big), \quad l = 1, \ldots, L. \tag{18}$$

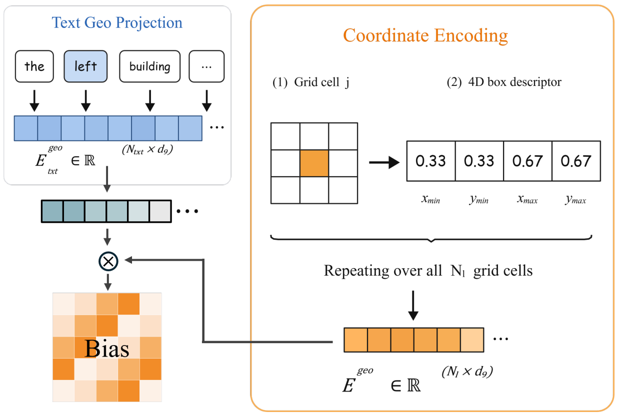

Figure 3: Internal mechanism of GSBI at one visual level. Contextual text tokens and visual tokens produce standard semantic logits, while normalized grid boxes are encoded and matched with projected text tokens to form a compact coordinate compatibility matrix. A factorized, input-adaptive per-head coefficient injects this matrix into the pre-softmax logits. The resulting attention output modulates the text stream and is passed to the next visual level.

The normalized coordinate representation makes parameter sharing well-defined across feature resolutions, while the recursion exposes the text stream successively to fine spatial detail and coarser scene context.

Fusion is asymmetric. The text-to-visual direction uses GSBI and propagates the updated linguistic representation across levels. The visual-to-text direction retains the baseline cross-attention module and always uses the initial text features $\mathbf{H}_{\text{txt}}^{(0)}$ as keys and values:

$$\widetilde{\mathbf{H}}_{\text{vis}}^{(l)} = \text{CrossAttn}\left(\mathbf{H}_{\text{vis}}^{(l)}, \mathbf{H}_{\text{txt}}^{(0)}\right). \tag{19}$$

This preserves a fixed semantic reference for each visual level while allowing only the text stream to undergo recurrent multi-scale refinement.

Parameter and Computational Overhead

GSBI replaces the original text-to-visual attention but retains projections of the same dimensionality for $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$, and the output. Relative to standard cross-attention, its additional parameters arise only from the coordinate pathway and adaptive gate. With model width d, geometry width d_{geo}, and H heads, the exact increment is

$$\Delta P = d_{\text{geo}}^2 + (d+7)d_{\text{geo}} + 2d^2 + d + dH + 2H. \tag{20}$$

For $d = 256$, $d_{\text{geo}} = 64$, and $H = 8$, Eq. 20 gives 154,320 parameters. The additional computation at level l is

$$\mathcal{O}\left(N_l d_{\text{geo}}^2 + M d d_{\text{geo}} + M N_l d_{\text{geo}} + d^2\right). \tag{21}$$

Because $\mathbf{B}_{\text{geo}}^{(l)}$ is shared across heads, its $M N_l d_{\text{geo}}$ product is computed once rather than H times. GSBI nevertheless retains dense token-to-region attention and therefore does not change its asymptotic dense-attention nature.

Text Encoding and Grounding Decoder

We use a two-layer bidirectional LSTM with hidden size 128, embedding dimension 300, and dropout 0.1. Its token-level outputs are projected to $d = 256$ and refined through Eq. 18. A learned pooling module aggregates $\mathbf{H}_{\text{txt}}^{(L)}$ into a sentence representation $\mathbf{h}_{\text{sent}}$.

The fused multi-scale visual features and $\mathbf{h}_{\text{sent}}$ are passed to a Deformable DETR-style decoder with Q object queries and D decoder layers. Each layer predicts a target confidence and a normalized box. Hungarian matching assigns predictions to the ground-truth target. The training objective combines focal, ℓ_1, and generalized IoU (GIoU) losses (Rezatofighi et al. 2019):

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{focal}} + \lambda_{L1}\|\mathbf{b} - \hat{\mathbf{b}}\|_1 + \lambda_{\text{giou}}\left(1 - \text{GIoU}(\mathbf{b}, \hat{\mathbf{b}})\right), \tag{22}$$

where $\lambda_{L1} = 5$ and $\lambda_{\text{giou}} = 2$. The same loss is applied to intermediate decoder layers for auxiliary supervision. The geometry pathway receives supervision solely through Eq. 22; no spatial-relation annotations or auxiliary geometric losses are introduced.

Experiments

Experimental Settings

Datasets. We evaluate GADAN on DIOR-RSVG (Zhan, Xiong, and Yuan 2023), containing 17,402 images and 38,320 expressions over 20 categories, and OPT-RSVG (Li et al. 2024), containing 25,452 images and 48,952 expressions over 14 categories.

Metrics. Following standard RSVG protocols, we report Precision@k (Pr@k) for $k \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$, mean IoU (mIoU), and cumulative IoU (cIoU).

Implementation. Images are resized to 640×640. Models are trained for 70 epochs using AdamW. The learning rates are 10^{-4} for the transformer, 5×10^{-5} for the visual backbone, and 10^{-5} for the text encoder; the weight decay is 5×10^{-4}, and the learning rate is decayed by 0.1 at epoch 60. Unless otherwise specified, $d_{\text{geo}} = 64$, the attention uses eight heads, the decoder has four layers, and ten object queries are used. In the controlled fusion–encoder comparison, LQVG and GSBI use either the same BiLSTM configuration described above or the same fine-tuned RoBERTa-base checkpoint, with the remaining training settings held fixed. Experiments are conducted on four RTX 3090 GPUs.

Comparison with Existing Methods

Table 1 summarizes the main results. On DIOR-RSVG, GADAN improves over LQVG, the most directly comparable published method with the same ResNet-50 backbone, by 2.09, 2.47, 2.60, 2.99, and 4.59 percentage points at Pr@0.5–Pr@0.9, respectively, together with a 2.41-point mIoU gain. Because this system-level comparison also changes the text encoder, we isolate the contribution of GSBI through encoder-matched controls in Table 2. The increasing improvement under stricter IoU thresholds is consistent with the intended role of GSBI: coordinate-conditioned attention helps discriminate semantically similar regions and yields

Dataset	Method	Venue	Vis.Enc	Text	Params	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	mIoU	cIoU
DIOR-RSVG	TransVG	CVPR'21	R-50	BERT	149.7	66.18	61.18	53.58	41.11	18.05	57.25	69.48
	LBYL-Net	CVPR'21	D-53	BERT	163.8	67.47	61.84	54.21	39.89	16.24	58.33	70.62
	QRNet	CVPR'22	Swin-S	BERT	178.4	69.77	63.31	54.57	42.82	18.66	59.05	71.69
	VLTVG	CVPR'22	R-50	BERT	152.2	69.13	63.36	55.05	43.65	19.10	59.69	72.05
	MGVLF	TGRS'23	R-50	BERT	152.5	70.07	64.43	55.89	43.40	21.32	60.64	74.40
	LQVG	TGRS'24	R-50	BERT†	166.3	81.81	79.40	74.31	63.24	40.85	72.70	80.90
	KEVG	JAG'25	Swin-L	BERT	—	82.31	77.01	69.81	57.47	37.51	71.06	79.42
	CSDNet	IF'25	R-101	BERT	—	81.91	—	71.05	—	32.04	70.88	79.58
	GADAN	—	R-50	BiLSTM	58.0	**83.90**	**81.87**	**76.91**	**66.23**	**45.44**	**75.11**	**82.93**
OPT-RSVG	TransVG	CVPR'21	R-50	BERT	149.7	69.96	64.17	54.68	38.01	12.75	59.80	69.31
	LBYL-Net	CVPR'21	D-53	BERT	163.8	70.22	65.39	58.65	37.54	9.46	60.57	70.28
	QRNet	CVPR'22	Swin-S	BERT	178.4	72.03	65.94	56.90	40.70	13.35	60.82	75.39
	VLTVG	CVPR'22	R-50	BERT	152.2	71.84	66.54	57.79	41.63	14.62	60.78	70.69
	VLTVG	CVPR'22	R-101	BERT	171.2	73.50	68.13	59.93	43.45	15.31	62.48	73.86
	MGVLF	TGRS'23	R-50	BERT	152.5	72.19	66.86	58.02	42.51	15.30	61.51	71.80
	LQVG	TGRS'24	R-50	BERT†	166.3	78.58	75.26	69.03	55.21	**26.83**	67.25	75.59
	LPVA	TGRS'24	R-50	BERT	–	78.03	73.32	62.22	49.60	25.61	66.20	76.30
	SegVG	ECCV'24	R-50	BERT	155.3	77.42	72.70	62.53	48.78	19.27	65.16	76.69
	GADAN	—	R-50	BiLSTM	58.0	**82.12**	**79.59**	**73.07**	**57.83**	24.96	**70.24**	**78.12**

Table 1: Comparison on DIOR-RSVG and OPT-RSVG. Vis.Enc denotes the visual encoder; Params are reported in millions. Bold values indicate the best result among the methods included in the table. The OPT-RSVG results for LQVG are taken from TIDM (Hu, Yang, and Li 2026). †Following the naming convention in prior RSVG comparison tables, LQVG's text encoder is labeled as BERT; its released implementation uses RoBERTa-base.

Fusion	Text Enc.	@0.5	@0.7	@0.9	mIoU	cIoU
LQVG	RoBERTa-base	81.81	74.31	40.85	72.70	80.90
LQVG	BiLSTM	82.13	74.21	41.31	72.82	81.32
GSBI	RoBERTa-base	83.71	76.68	43.99	74.39	82.63
GSBI	BiLSTM	**83.90**	**76.91**	**45.44**	**75.11**	**82.93**
GSBI	BERT-tiny	83.11	75.96	43.43	73.98	82.48
GSBI	RoBERTa-base*	82.88	76.17	44.03	73.96	82.50

Table 2: Fusion and text encoder comparison on DIOR-RSVG. The first four rows form a 2×2 comparison between LQVG fusion and GSBI under RoBERTa-base and BiLSTM. The final two rows provide encoder-capacity controls. Pretrained encoders are fine-tuned unless marked * (frozen).

more precise target selection. On OPT-RSVG, GADAN surpasses LPVA by 4.04 mIoU points. Compared with LQVG, it gains 3.54, 4.33, 4.04, and 2.62 points at Pr@0.5–Pr@0.8, respectively, while improving mIoU and cIoU by 2.99 and 2.53 points. Its Pr@0.9 is 1.87 points lower than LQVG, showing that the improvement does not extend to the strictest overlap threshold. These results demonstrate strong cross-dataset performance rather than establishing a universal state-of-the-art claim against methods not included in the table.

Controlled Fusion and Text Encoder Analysis

Table 2 separates the effect of the fusion module from that of the text encoder. With BiLSTM fixed, replacing LQVG fusion with GSBI raises Pr@0.5–Pr@0.9 from 82.13, 79.28, 74.21, 64.03, and 41.31 to 83.90, 81.87, 76.91, 66.23, and 45.44. The gains are 1.77, 2.59, 2.70, 2.20, and 4.13 points, while mIoU increases from 72.82 to 75.11 (+2.29) and cIoU from 81.32 to 82.93 (+1.61). With RoBERTa-base fixed, GSBI improves Pr@0.5, Pr@0.7, Pr@0.9, mIoU, and cIoU by 1.90, 2.37, 3.14, 1.69, and 1.73 points. These encoder-matched comparisons show that the benefit of GSBI is not attributable to changing the text encoder.

Without GSBI, BiLSTM and RoBERTa-base obtain similar mIoU values (72.82 versus 72.70); with GSBI, the corresponding results are 75.11 and 74.39. The resulting difference-in-differences is 0.60 mIoU, suggesting descriptively that GSBI may benefit the BiLSTM setting somewhat more. Because all results are from single runs, we do not interpret this contrast as evidence of a statistically reliable interaction. The appropriate conclusion is that GSBI consistently improves both encoders, while BiLSTM provides the strongest final accuracy under the present training protocol.

Viewed as a factorial comparison, averaging over text encoders gives 74.75 mIoU for GSBI and 72.76 for LQVG fusion, corresponding to a 1.99-point fusion effect. Averaging over fusion variants gives 73.97 for BiLSTM and 73.55 for RoBERTa-base, corresponding to only a 0.42-point encoder effect. These averages reinforce the encoder-matched comparisons: most of the performance change is associated with coordinate-conditioned fusion rather than with replacing RoBERTa-base by BiLSTM. Under BiLSTM, the GSBI gain becomes largest at the strictest threshold, reaching 4.13 points at Pr@0.9 compared with 1.77 points at Pr@0.5. Since the decoder and regression objective remain unchanged, this

Table 3: Keyword-level diagnostic results on DIOR-RSVG. All values are reported in percentages, and Δ denotes the difference between GADAN and LQVG.

(a) mIoU

Keyword	Samples	LQVG	GADAN	Δ
right	1,980	71.65	74.69	+3.04
left	1,925	70.54	74.21	+3.67
middle	1,403	80.17	75.23	−4.94
lower	1,111	70.98	74.80	+3.82
upper	1,059	69.57	73.89	+4.32
bottom	626	73.13	71.93	−1.20
top	599	72.60	75.33	+2.73
lower left	565	69.90	74.35	+4.45
lower right	552	71.89	75.02	+3.13
upper left	546	67.98	74.30	+6.32
upper right	528	70.43	73.31	+2.88
above	158	57.37	70.10	+12.73
below	156	62.36	73.63	+11.27

(b) Pr@0.7

Keyword	Samples	LQVG	GADAN	Δ
right	1,980	73.48	76.41	+2.93
left	1,925	72.31	76.57	+4.26
middle	1,403	81.47	76.84	−4.63
lower	1,111	73.36	77.05	+3.69
upper	1,059	70.54	76.39	+5.85
bottom	626	75.88	74.28	−1.60
top	599	73.62	78.63	+5.01
lower left	565	72.92	77.52	+4.60
lower right	552	73.55	76.27	+2.72
upper left	546	68.86	76.37	+7.51
upper right	528	71.40	76.14	+4.74
above	158	58.86	70.25	+11.39
below	156	62.82	76.28	+13.46

Note. Keywords are matched independently. A single expression may therefore contribute to multiple subsets, and the sample counts are not additive.

pattern is consistent with better spatially conditioned fused features rather than a new box-regression mechanism.

Keyword-Level Spatial Analysis

We conduct a keyword-level diagnostic on DIOR-RSVG, evaluating frequent directional and relational terms. Because keywords are matched independently, the subsets in Table 3 may overlap.

GADAN improves mIoU and Pr@0.7 on 11 of 13 subsets. The largest gains occur for "above" (+12.73 mIoU and +11.39 Pr@0.7 points) and "below" (+11.27 and +13.46 points), while compound and individual directional terms generally improve. These results indicate that coordinate evidence complements contextual visual–language features for direction-sensitive queries.

Performance declines for "middle" (−4.94 mIoU and −4.63 Pr@0.7 points) and slightly for "bottom" (−1.20 and −1.60 points). These terms may express ordinal or object-relative positions rather than global image regions, which normalized absolute coordinates cannot fully capture. Thus, GSBI is most effective when spatial terms correspond stably to image coordinates, while inter-instance ordering and local geometric reasoning remain challenging.

Limitations

The coordinate pathway uses normalized grid boxes and provides its most direct inductive bias for image-centric absolute positions and coarse spatial regions. It does not explicitly encode pairwise geometry between detected objects. Relational expressions may still benefit from contextual visual-language features, but explicit object-to-object reasoning is a natural direction for future work. Moreover, the modest differences among text encoders and the descriptive fusion–encoder interaction should be validated with repeated runs before making claims about optimization stability or representation structure. Although the BiLSTM configuration reduces the parameter count, GSBI retains dense token-to-region attention and does not change the overall FLOP profile; we therefore do not claim computational efficiency.

Because GSBI uses image-normalized coordinates, its robustness to rotations, flips, crops, and changes in image orientation remains untested. Rotation-invariant remote-sensing transformers suggest that handling arbitrary orientations can require dedicated architectural treatment (Sun et al. 2024), which GSBI currently lacks. Improvements on absolute directional terms may partly reflect benchmark-specific spatial regularities. Geometric perturbations and cross-region evaluation would help distinguish transferable language-conditioned geometry from dataset-specific position priors.

Conclusion

Normalized grid boxes primarily encode image-centric positions and coarse regions, not pairwise object geometry, whereas prior grounding models explicitly exploit inter-object relationships (Yu et al. 2018); explicit object-to-object reasoning remains future work. The modest encoder differences and descriptive fusion–encoder interaction require repeated runs before claims about stability or representation structure. Although BiLSTM reduces parameters, GSBI retains dense token-to-region attention and does not reduce the FLOP profile, so we do not claim computational efficiency.

GSBI's robustness to rotations, flips, crops, and orientation changes remains untested; rotation-invariant RS transformers indicate that arbitrary orientations may require dedicated treatment (Sun et al. 2024). Geometric perturbations and cross-region evaluation are needed to separate transferable geometry from dataset-specific position priors.

References

Cao, X.; Yao, J.; Xu, Z.; and Meng, D. 2020. Hyperspectral Image Classification with Convolutional Neural Network and Active Learning. *IEEE Transactions on Geoscience and Remote Sensing*, 58(7): 4604–4616.

Cao, X.; Zhou, F.; Xu, L.; Meng, D.; Xu, Z.; and Paisley, J. 2018. Hyperspectral Image Classification with Markov

Random Fields and a Convolutional Neural Network. *IEEE Transactions on Image Processing*, 27(5): 2354–2367.

Carion, N.; Massa, F.; Synnaeve, G.; Usunier, N.; Kirillov, A.; and Zagoruyko, S. 2020. End-to-End Object Detection with Transformers. In *Proceedings of the European Conference on Computer Vision*, 213–229.

Deng, J.; Yang, Z.; Chen, T.; Zhou, W.; and Li, H. 2021. TransVG: End-to-End Visual Grounding with Transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 1769–1779.

Ding, J.; Xue, N.; Xia, G.-S.; Bai, X.; Yang, W.; Yang, M. Y.; Belongie, S.; Luo, J.; Datcu, M.; Pelillo, M.; and Zhang, L. 2022a. Object Detection in Aerial Images: A Large-Scale Benchmark and Challenges. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(11): 7778–7796.

Ding, J.; Xue, N.; Xia, G.-S.; and Dai, D. 2022b. Decoupling zero-shot semantic segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 11583–11592.

Hu, F.; Xia, G.-S.; Hu, J.; and Zhang, L. 2015. Transferring Deep Convolutional Neural Networks for the Scene Classification of High-Resolution Remote Sensing Imagery. *Remote Sensing*, 7(11): 14680–14707.

Hu, M.; Yang, K.; and Li, J. 2026. Text-Injected Discriminative Model for Remote Sensing Visual Grounding. *Remote Sensing*, 18(1): 161.

Hu, R.; Xu, H.; Rohrbach, M.; Feng, J.; Saenko, K.; and Darrell, T. 2016. Natural Language Object Retrieval. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 4555–4564.

Huang, B.; Lian, D.; Luo, W.; and Gao, S. 2021. Look before you leap: Learning landmark features for one-stage visual grounding. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 16888–16897.

Kamath, A.; Singh, M.; LeCun, Y.; Synnaeve, G.; Misra, I.; and Carion, N. 2021. MDETR: Modulated Detection for End-to-End Multi-Modal Understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 1780–1790.

Kang, W.; Liu, G.; Shah, M.; and Yan, Y. 2024. Segvg: Transferring object bounding box to segmentation for visual grounding. In *European Conference on Computer Vision*, 57–75. Springer.

Kuckreja, K.; Danish, M. S.; Naseer, M.; Das, A.; Khan, S.; and Khan, F. S. 2024. GeoChat: Grounded Large Vision-Language Model for Remote Sensing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 27831–27840.

Lan, M.; Rong, F.; Jiao, H.; Gao, Z.; and Zhang, L. 2024. Language Query-Based Transformer with Multiscale Cross-Modal Alignment for Visual Grounding on Remote Sensing Images. *IEEE Transactions on Geoscience and Remote Sensing*, 62: 1–13.

Li, K.; Wan, G.; Cheng, G.; Meng, L.; and Han, J. 2020. Object Detection in Optical Remote Sensing Images: A Survey and a New Benchmark. *ISPRS Journal of Photogrammetry and Remote Sensing*, 159: 296–307.

Li, K.; Wang, D.; Xu, H.; Zhong, H.; and Wang, C. 2024. Language-Guided Progressive Attention for Visual Grounding in Remote Sensing Images. *IEEE Transactions on Geoscience and Remote Sensing*, 62: 1–13.

Li, L. H.; Zhang, P.; Zhang, H.; Yang, J.; Li, C.; Zhong, Y.; Wang, L.; Yuan, L.; Zhang, L.; Hwang, J.-N.; Chang, K.-W.; and Gao, J. 2022. Grounded Language-Image Pre-Training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 10965–10975.

Liu, F.; Chen, D.; Guan, Z.; Zhou, X.; Zhu, J.; Ye, Q.; Fu, L.; and Zhou, J. 2024a. RemoteCLIP: A Vision Language Foundation Model for Remote Sensing. *IEEE Transactions on Geoscience and Remote Sensing*, 62: 1–16.

Liu, S.; Ma, Y.; Zhang, X.; Wang, H.; Ji, J.; Sun, X.; and Ji, R. 2024b. Rotated Multi-Scale Interaction Network for Referring Remote Sensing Image Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 26658–26668.

Liu, S.; Zeng, Z.; Ren, T.; Li, F.; Zhang, H.; Yang, J.; Jiang, Q.; Li, C.; Yang, J.; Su, H.; Zhu, J.; and Zhang, L. 2024c. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. In *Proceedings of the European Conference on Computer Vision*, 38–55.

Liu, Z.; Lin, Y.; Cao, Y.; Hu, H.; Wei, Y.; Zhang, Z.; Lin, S.; and Guo, B. 2021. Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 10012–10022.

Luo, F.; Du, B.; Zhang, L.; Zhang, L.; and Tao, D. 2019. Feature Learning Using Spatial–Spectral Hypergraph Discriminant Analysis for Hyperspectral Image. *IEEE Transactions on Cybernetics*, 49(7): 2406–2419.

Ma, Q.; Li, L.; Lu, X.; Jiao, L.; Liu, F.; Ma, W.; Liu, X.; and Sun, L. 2025. LSCF: Long-Term Semantic-Guidance ConvFormer for Referring Remote Sensing Image Segmentation. *IEEE Transactions on Geoscience and Remote Sensing*, 63: 1–13.

Raffel, C.; Shazeer, N.; Roberts, A.; Lee, K.; Narang, S.; Matena, M.; Zhou, Y.; Li, W.; and Liu, P. J. 2020. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *Journal of Machine Learning Research*, 21(140): 1–67.

Ren, S.; He, K.; Girshick, R.; and Sun, J. 2017. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(6): 1137–1149.

Rezatofighi, H.; Tsoi, N.; Gwak, J.; Sadeghian, A.; Reid, I.; and Savarese, S. 2019. Generalized Intersection over Union: A Metric and a Loss for Bounding Box Regression. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 658–666.

Shaw, P.; Uszkoreit, J.; and Vaswani, A. 2018. Self-Attention with Relative Position Representations. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 464–468.

She, K.; Zhang, M.; Zhao, Y.; Yang, B.; and Liu, Q. 2025. From Object to Context: Scene Knowledge Enhanced Visual Grounding for Geospatial Understanding. *International Journal of Applied Earth Observation and Geoinformation*, 142: 104706.

Sun, L.; Li, C.; Jiao, L.; Li, L.; Liu, X.; Liu, F.; Yang, S.; and Hou, B. 2024. Lighter and Robust: A Rotation-Invariant Transformer for VHR Image Change Detection. *IEEE Transactions on Geoscience and Remote Sensing*, 62: 1–14.

Sun, Y.; Feng, S.; Li, X.; Ye, Y.; Kang, J.; and Huang, X. 2022. Visual Grounding in Remote Sensing Images. In *Proceedings of the 30th ACM International Conference on Multimedia*, 404–412.

Tong, X.-Y.; Xia, G.-S.; Lu, Q.; Shen, H.; Li, S.; You, S.; and Zhang, L. 2020. Land-Cover Classification with High-Resolution Remote Sensing Images Using Transferable Deep Models. *Remote Sensing of Environment*, 237: 111322.

Wan, S.; Gong, C.; Zhong, P.; Du, B.; Zhang, L.; and Yang, J. 2020. Multiscale Dynamic Graph Convolutional Network for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing*, 58(5): 3162–3177.

Xia, G.-S.; Bai, X.; Ding, J.; Zhu, Z.; Belongie, S.; Luo, J.; Datcu, M.; Pelillo, M.; and Zhang, L. 2018. DOTA: A Large-Scale Dataset for Object Detection in Aerial Images. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 3974–3983.

Xia, G.-S.; Hu, J.; Hu, F.; Shi, B.; Bai, X.; Zhong, Y.; Zhang, L.; and Lu, X. 2017. AID: A Benchmark Data Set for Performance Evaluation of Aerial Scene Classification. *IEEE Transactions on Geoscience and Remote Sensing*, 55(7): 3965–3981.

Xu, C.; Wang, J.; Yang, W.; Yu, H.; Yu, L.; and Xia, G.-S. 2022. Detecting Tiny Objects in Aerial Images: A Normalized Wasserstein Distance and a New Benchmark. *ISPRS Journal of Photogrammetry and Remote Sensing*, 190: 79–93.

Yang, L.; Xu, Y.; Yuan, C.; Liu, W.; Li, B.; and Hu, W. 2022. Improving Visual Grounding with Visual-Linguistic Verification and Iterative Reasoning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 9499–9508.

Yang, Z.; Gong, B.; Wang, L.; Huang, W.; Yu, D.; and Luo, J. 2019. A Fast and Accurate One-Stage Approach to Visual Grounding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 4683–4693.

Ye, J.; Tian, J.; Yan, M.; Yang, X.; Wang, X.; Zhang, J.; He, L.; and Lin, X. 2022. Shifting More Attention to Visual Backbone: Query-Modulated Refinement Networks for End-to-End Visual Grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 15502–15512.

Yu, L.; Lin, Z.; Shen, X.; Yang, J.; Lu, X.; Bansal, M.; and Berg, T. L. 2018. MAttNet: Modular Attention Network for Referring Expression Comprehension. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 1307–1315.

Yu, L.; Poirson, P.; Yang, S.; Berg, A. C.; and Berg, T. L. 2016. Modeling Context in Referring Expressions. In *Proceedings of the European Conference on Computer Vision*, 69–85.

Yuan, Z.; Mou, L.; Hua, Y.; and Zhu, X. X. 2024. RRSIS: Referring Remote Sensing Image Segmentation. *IEEE Transactions on Geoscience and Remote Sensing*, 62: 1–12.

Zhan, Y.; Xiong, Z.; and Yuan, Y. 2023. RSVG: Exploring Data and Models for Visual Grounding on Remote Sensing Data. *IEEE Transactions on Geoscience and Remote Sensing*, 61: 1–13.

Zhang, L.; Zhang, L.; and Du, B. 2016. Deep Learning for Remote Sensing Data: A Technical Tutorial on the State of the Art. *IEEE Geoscience and Remote Sensing Magazine*, 4(2): 22–40.

Zhang, W.; Cai, M.; Zhang, T.; Zhuang, Y.; and Mao, X. 2024a. EarthGPT: A Universal Multimodal Large Language Model for Multisensor Image Comprehension in Remote Sensing Domain. *IEEE Transactions on Geoscience and Remote Sensing*, 62: 1–20.

Zhang, Z.; Zhao, T.; Guo, Y.; and Yin, J. 2024b. RS5M and GeoRSCLIP: A Large-Scale Vision-Language Dataset and a Large Vision-Language Model for Remote Sensing. *IEEE Transactions on Geoscience and Remote Sensing*, 62: 1–23.

Zhao, E.; Wan, Z.; Zhang, Z.; Nie, J.; Liang, X.; and Huang, L. 2024. A Spatial-Frequency Fusion Strategy Based on Linguistic Query Refinement for RSVG. *IEEE Transactions on Geoscience and Remote Sensing*, 62: 1–13.

Zhao, Y.; Chen, Y.; Yao, R.; Xiong, S.; and Lu, X. 2025. Context-Driven and Sparse Decoding for Remote Sensing Visual Grounding. *Information Fusion*, 123: 103296.

Zhu, X. X.; Tuia, D.; Mou, L.; Xia, G.-S.; Zhang, L.; Xu, F.; and Fraundorfer, F. 2017. Deep Learning in Remote Sensing: A Comprehensive Review and List of Resources. *IEEE Geoscience and Remote Sensing Magazine*, 5(4): 8–36.